

Physics-Informed Neural Networks for the Poisson Equation

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ABSTRACT. We study physics-informed neural networks (PINNs) for solving the Poisson equation on a bounded domain $\Omega \subset \mathbb{R}^2$. The network u_θ is trained to satisfy the PDE residual together with the boundary data, removing the need for a meshed solver.

2. Method

We seek u_θ approximating the solution of the Poisson problem:

$$-\Delta u(x) = f(x), \quad x \in \Omega, \quad u(x) = g(x), \quad x \in \partial\Omega. \quad (1)$$

The composite training objective combines a data term and a PDE residual:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{data}}(\theta) + \lambda \mathcal{L}_{\text{pde}}(\theta), \quad (2)$$

where the residual is evaluated at N_r collocation points:

$$\mathcal{L}_{\text{pde}}(\theta) = \frac{1}{N_r} \sum_{i=1}^{N_r} |\Delta u_\theta(x_i) + f(x_i)|^2. \quad (3)$$

Minimizing $\mathcal{L}(\theta)$ drives the network toward a solution that is consistent with both the observed data and the governing equation, while the weight λ balances the two terms.